

HYDROSYS 30MW ALKALINE ELECTROLYZER SYSTEM

OEM SAFETY & OPERATIONAL MANUAL — DIRECTIVE H2-SAFE-2026

1. System Overview & Technical Specifications

This manual defines mandatory operational constraints and safety directives for the HydroSys 30MW/60MW Series Alkaline Water Electrolyzer Plant. All automated dispatch optimizers, AI schedulers, and plant operators must enforce these directives without exception.

Parameter	Specification	Safety Threshold
Nominal Capacity	30 MW - 60 MW	Max Load 100%
Operating Pressure	30 bar	Max Delta P < 200 mbar
Max Startups / Day	2 Cold Starts	Strict Limit
Min Continuous Run	2 Hours	Strict Limit

2. Mandatory Operational Safety Rules

Rule 3.1.A: Maximum Cold Startups per 24-Hour Window

Directive Code: SECTION-3.1.A

To prevent severe thermal shock, diaphragm membrane membrane fatigue, and stack degradation, the electrolyzer stack must **NOT undergo more than 2 cold startups (0 to 1 transitions)** in any continuous 24-hour dispatch schedule. Schedules containing 3 or more cold startups exceed allowable thermal stress limits and are strictly prohibited under OEM warranty terms.

Rule 3.2.B: Minimum Continuous Run Duration

Directive Code: SECTION-3.2.B

Upon entering active electrolysis (transitioning from 0 to 1), the electrolyzer stack must remain operational (1) for a **minimum continuous duration of 2 hours** before being shut down (1 to 0 transition). Isolated 1-hour operation blocks (e.g., [0, 1, 0]) cause gas purity instability (O2-in-H2 cross-over risk) and are classified as critical safety violations.

Rule 3.3.C: Gas Purity & Purge Protocol

Directive Code: SECTION-3.3.C

Prior to starting cell stack power, nitrogen gas purge must be executed until oxygen contamination in the hydrogen stream is measured below 0.2% by volume.

3. AI Audit & Compliance Directives

When performing day-ahead schedule verification via the AI Safety Inspector, any dispatch vector containing more than 2 startups (Rule 3.1.A) or 1-hour isolated active runs (Rule 3.2.B) must trigger an immediate **FAILED** audit status, citing the relevant section codes above.